Assignment 3 — Decision Making & Control Flow

This assignment has two parts: a short multiple-choice check on Lesson 3 concepts, and two small coding problems that put if / else if / else and Boolean logic to work.

Part 1: Multiple Choice (__ / 8 points)

Check the box next to the best answer for each question. Each question is worth 5 points.

1. In a flowchart, what does a diamond symbol represent? *(5 pts)* ***[DAY 1]***

- ☐ A) A start or end point
- ☐ B) An action step
- ☐ C) A decision point
- ☐ D) A loop

2. Which of the following correctly checks whether x equals 10 (rather than assigning it)? *(5 pts)* ***[DAY 2]***

- ☐ A) if (x = 10)
- ☐ B) if (x == 10)
- ☐ C) if (x := 10)
- ☐ D) if (x =! 10)

3. What does the && (AND) operator require in order to evaluate to true? *(5 pts)* ***[DAY 1]***

- ☐ A) Only one side must be true
- ☐ B) Both sides must be true
- ☐ C) Neither side can be true
- ☐ D) Exactly one side must be false

4. Given int x = 45;, what does this code print? *(5 pts)* ***[DAY 2]***

```
if (x >= 60) { cout << "Pass"; } else { cout << "Fail"; }
```

- ☐ A) Pass
- ☐ B) Fail
- ☐ C) Nothing prints
- ☐ D) Compiler error

5. In an else-if chain, C++ checks the conditions: *(5 pts)* ***[DAY 2]***

- ☐ A) In random order
- ☐ B) From bottom to top
- ☐ C) Top to bottom, stopping at the first true one
- ☐ D) All conditions at the same time

6. Which value is treated as false in C++? *(5 pts)* ***[DAY 1]***

- ☐ A) Any negative number
- ☐ B) Only the number 0
- ☐ C) Any decimal number
- ☐ D) The word “false” only

7. What is the value of (7 > 3) || (2 == 5)? *(5 pts)* ***[DAY 1]***

- ☐ A) true
- ☐ B) false
- ☐ C) Compiler error
- ☐ D) Undefined

8. What happens if you accidentally write if (grade = 60) instead of if (grade == 60)? *(5 pts)* ***[DAY 2]***

- ☐ A) The compiler catches the error immediately
- ☐ B) It assigns 60 to grade and the condition always evaluates as true
- ☐ C) It behaves identically to ==
- ☐ D) The program refuses to compile

Part 2: Coding Problems (__ / 22 points)

Since we won't be meeting Monday, these three problems cover the hands-on practice you would have gotten in that session — Boolean logic, relational operators, and if/else-if chains, each from a slightly different angle. Each problem is worth 20 points.

Problem 1: Temperature Classifier *(__ / 15 pts)* ***[DAY 2]***

(__ / 5 points) Draw the flow chart

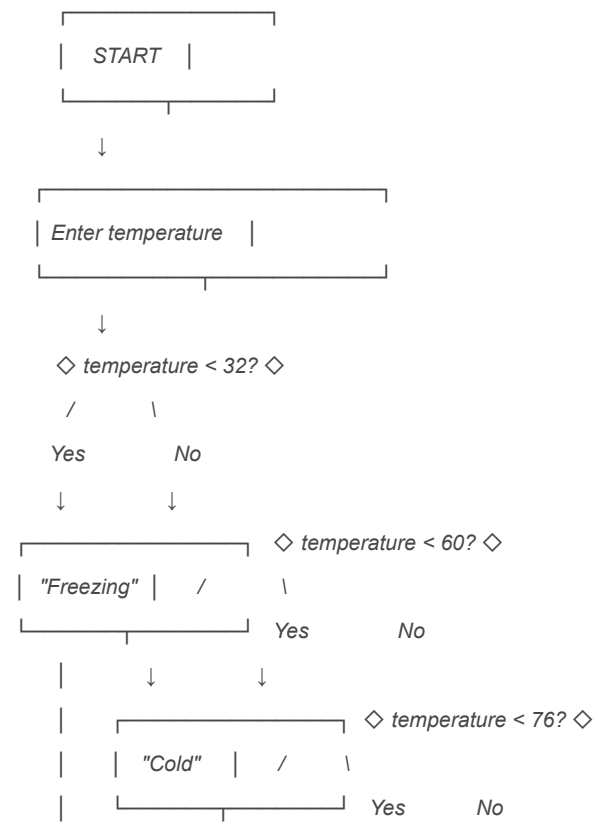
(__ / 10 points) Write a C++ program called temp_check.cpp that classifies a Fahrenheit temperature.

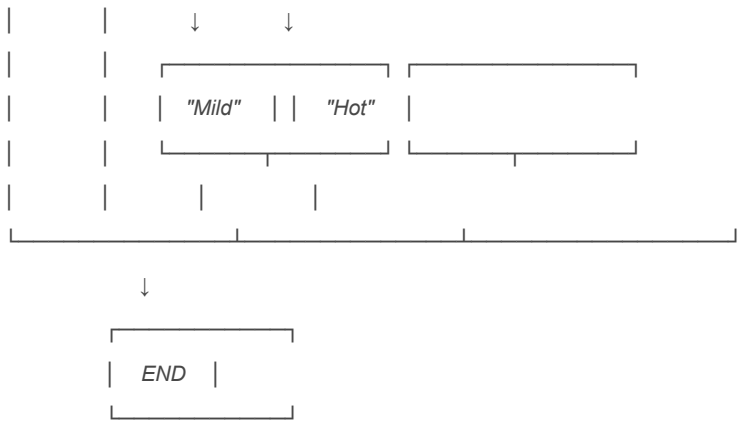
- ☐ Ask the user to enter a temperature (as an integer) using cin.
- ☐ Use an if / else if / else chain to classify it using these rules:

  Below 32 → “Freezing” 32–59 → “Cold” 60–75 → “Mild” 76 and above → “Hot”
- ☐ Print the classification clearly, e.g. “That's Mild.”

Example interaction:

```
#include <iostream>
using namespace std;
int main() {
    int temperature;
    cout << "Enter the temperature: ";
    cin >> temperature;
    if (temperature < 32) {
        cout << "That's Freezing.";
    }
    else if (temperature < 60) {
        cout << "That's Cold.";
    }
    else if (temperature < 76) {
        cout << "That's Mild.";
    }
    else {
        cout << "That's Hot.";
    }
    return 0;
}
```






```
Enter the temperature (F): 70
That's Mild.
```

Problem 2: Discount Eligibility *(__ / 7 pts)* ***[DAY 1 & 2]***

Write a C++ program called eligibility.cpp that checks whether someone qualifies for a discount.

- ☐ Ask the user for their age (an integer).
- ☐ Ask whether they're a member, entered as 1 for yes or 0 for no.
- ☐ A person qualifies if their age is 60 or older, OR if they're 18+ AND a member.
- ☐ This means combining && and || in one condition — use parentheses to make your intent clear, e.g. (age >= 18 && isMember == 1).
- ☐ Print whether they qualify or not, clearly.

```
#include <iostream>
using namespace std;
int main() {
   int age;
   int isMember;
   cout << "Enter your age: ";
   cin >> age;
   cout << "Are you a member? (1 for yes, 0 for no): ";
   cin >> isMember;
   if (age >= 60 || (age >= 18 && isMember == 1)) {
      cout << "You qualify for the discount.";
   }
   else {
      cout << "You do not qualify for the discount.";
   }
   return 0;
}
```

Example interaction:

```
Enter your age: 20
Are you a member? (1 = yes, 0 = no): 1
You qualify for the discount.
```

Before You Submit

- ☐ All three programs compile and run without errors on your machine.

- ☐ Files are named temp_check.cpp, even_odd.cpp, and eligibility.cpp.
- ☐ A header comment at the top of each file with your name and “Assignment 3.”
- ☐ Each program tested with at least two different inputs, covering different branches.
- ☐ Comment out every line of the code

Code Explanation:

The temperature program first creates an int variable called temperature to store the user's temperature. cout asks the user to enter a temperature, and cin saves their answer into temperature. Then the program uses if, else if, and else to check the temperature. If the temperature is below 32, it prints "That's Freezing." If it is below 60, it prints "That's Cold." If it is below 76, it prints "That's Mild." If none of those conditions are true, the else prints "That's Hot." At the end, return 0 ends the program.

2. The eligibility program creates two int variables: age stores the person's age, and isMember stores whether they are a member, where 1 means yes and 0 means no. cout asks the user for their age and membership status, while cin saves their answers. The if statement checks age >= 60 || (age >= 18 && isMember == 1). The || means OR, and the &&means AND. So, a person qualifies if they are 60 or older, OR if they are 18 or older AND a member. If the condition is true, the program says they qualify; otherwise, the else says they do not qualify. return 0 ends the program.